

UNSW School of Mathematics and Statistics

MATH2601 Higher Linear Algebra

Tutorial problems, Week 4

1. Let $T : V \rightarrow W$ be a linear map between two finite dimensional vector spaces.
 - a. Prove that if $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is a linearly independent set in W then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent in V .
 - b. Prove that if T is one-to-one and $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a linearly independent set in V then $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is a linearly independent set in W .
 - c. Prove that if T is onto and $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ spans V then $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ spans W .
 - d. If $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ spans W , must the set $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ span V ? Prove it must or give a counter-example.
2. If $A \in M_{p,q}(\mathbb{F})$ has $\text{rank}(A) = q$, prove that the linear transformation $T : \mathbb{F}^q \rightarrow \mathbb{F}^p$ defined by $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one.

What are the relative sizes of p and q ?

3. Let a, b, c be three distinct real constants, and consider the function $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ defined by

$$T(p) = (p(a), p'(b), p(c)) ,$$

where p' denotes the derivative of p .

- a. Prove that T is a linear transformation.
- b. Find the matrix of T with respect to the standard bases of $\mathcal{P}_2(\mathbb{R})$ and $\mathbb{R}^3$.
- c. Find conditions on a, b, c such that T is invertible. Can you explain your answer in terms of ideas from first-year (or earlier) calculus?
- d. Now take specifically $a = 1$, $b = 2$ and $c = 3$. Let M be the matrix of T with respect to the bases

$$\mathcal{B} = \{1 + t, 1 - t, 1 + t^2\} \quad \text{and} \quad \mathcal{C} = \{(2, 1, 4), (0, 1, 2), (1, 7, -3)\}$$

for $\mathcal{P}_2(\mathbb{R})$ and $\mathbb{R}^3$ respectively, i.e. $M = [T]_{\mathcal{C}}^{\mathcal{B}}$. Write down an expression for M as a product of three matrices; then see if you can evaluate M **without** calculating any matrix inverses. (If not, find the required inverse and evaluate M anyway.)

4. a. Consider the bases $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \end{pmatrix} \right\}$ and $\mathcal{C} = \left\{ \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\}$ for $\mathbb{R}^2$. If the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ has matrix

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 3 & 5 \\ 4 & 9 \end{pmatrix}$$

with respect to the basis $\mathcal{B}$ in the domain and $\mathcal{C}$ in the codomain, find the matrix of T with respect to standard bases of $\mathbb{R}^2$ in domain and codomain, i.e. find $[T]_{\mathcal{S}}^{\mathcal{S}}$ where $\mathcal{S}$ is the standard basis of $\mathbb{R}^2$. (**Note:** take care over the order of the products!)

- b. If a linear transformation $T : V \rightarrow W$ (where V and W are finite-dimensional) has matrix $A = [T]_{\mathcal{S}'}^{\mathcal{B}}$ with respect to a basis $\mathcal{B}$ for V and the standard basis $\mathcal{S}'$ for W , find an expression for the matrix of T with respect to the standard basis $\mathcal{S}$ for V and a basis $\mathcal{C}$ for W , i.e. find $[T]_{\mathcal{C}}^{\mathcal{S}}$.
5. Let V be the real vector space discussed in lectures where the underlying set is $\mathbb{R}^+$, the positive reals, with addition and scalar multiplication respectively defined on V by

$$\mathbf{v} \oplus \mathbf{w} = \mathbf{vw} \quad \text{and} \quad \alpha \otimes \mathbf{v} = \mathbf{v}^{\alpha}.$$

For any fixed $t \in \mathbb{R}$, define $T : V \rightarrow V$ to be the map $T(x) = x^t$.

- a. Prove that $T \in L(V, V)$.
- b. Choosing any basis $\mathcal{B}$ of V , find the matrix of T with respect to that basis, $[T]_{\mathcal{B}}^{\mathcal{B}}$, and convince yourself that the usual rule applies:

$$[T(x)]_{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}}[x]_{\mathcal{B}}$$

where the product on the right is ordinary matrix multiplication.